

OPPONG KWABENA ADUTWUM

MACHINE LEARNING ENGINEER

Applied ML · Robotics · Embedded Intelligence · Full-Stack Development

CONTACT

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TECHNICAL SKILLS

Programming Languages

- Python
- JavaScript
- C++
- HTML / CSS

Data & Analytics

- Pandas / NumPy
- SQL / PostgreSQL
- Jupyter

AI & LLM Tools

- Scikit-learn
- Groq AI / Anthropic Claude API
- LLM Prompt Chaining / RAG

Web & Backend

- FastAPI / Flask
- React Native / Django
- SQLite (offline-first)

Dev Tools & Design

- Git / GitHub
- VS Code / CLion
- Figma
- Linux (Kali)

Core Knowledge

- Machine Learning (fundamentals)
- Network Security
- Embedded Systems / IoT
- PID Control / Robotics
- OSI & TCP/IP Networking

LANGUAGES

- English — Native
- French — Intermediate

CERTIFICATIONS

- Hashgraph Developer Course — Hedera (Jan 2026)
- Hedera Certified Foundation (Apr 2026)
- IPPF Competitor — Brewer Foundation & NYU (2024-25)
- WASSCE General Science — Prempeh College (Sep 2025)
- [Deep Learning Spec.] — DeepLearning.AI
- [ML Specialization] — Stanford / Coursera

GITHUB HIGHLIGHTS

- [repo] — CNN crop disease classifier ★[X]
- [repo] — LLM-powered study assistant ★[X]
- [repo] — Sentiment analysis pipeline ★[X]
- [X] contributions | [X] PRs (12 mo.)

OPEN SOURCE

- [Hugging Face / PyTorch] — PR #[X]
- [scikit-learn / LangChain] — PR #[X]

RESEARCH & BLOG

- "[Title]" — [arXiv / Conference]
- "[Title]" — [Medium / Hashnode]

INTERESTS

- Mathematics

PROFESSIONAL SUMMARY

Computer Science undergraduate at KNUST with hands-on experience in embedded intelligence, autonomous systems, and applied machine learning. National Robotics Champion. Regional Cybersecurity Champion.

EXPERIENCE

Machine Learning Trainee — Zap Tech Technologies

Remote | May 2025

- Completed an intensive 2-week ML engineering program covering the full model development lifecycle — from raw data collection through model evaluation and iteration.
- Preprocessed and cleaned real-world datasets using Pandas and NumPy, handling missing values, feature scaling, and data augmentation.
- Trained and evaluated classification models using Scikit-learn, interpreting accuracy, precision, recall, and F1-score across iterations.

Co-Founder & Software Developer — Arethos Tech Initiatives

Kumasi, Ghana | 2026 - Present

- Co-built Arethos, a cross-platform mental wellness app for KNUST students: AI-powered journaling, mood tracking, and reflections (React Native, Expo SDK 54, SQLite, Groq + Anthropic AI dual-fallback).
- Engineered custom haptic feedback (9 iterations), offline-first SQLite storage, and high-fidelity Figma component design systems; coordinating sprint tasks via Git.
- Leading the ACES CodeFest 2026 UI/UX Challenge entry — redesigning the ACES website into a mobile-first experience across 9 pages in a 2-week sprint.

[AI/ML Engineering Intern] — [Company Name — e.g. Anthropic, Google DeepMind]

[City, Country] | [Mon YYYY - Mon YYYY]

- Trained and fine-tuned [model type] on [dataset], improving [metric] by [X]% over baseline.
- Built end-to-end ML pipeline: data ingestion → preprocessing → training → evaluation → deployment on [AWS/GCP].

[ML Research Assistant] — [KNUST AI Lab / Dept.]

Kumasi, Ghana | [Mon YYYY - Present]

- Researched [topic, e.g. NLP for low-resource African languages] under Prof. [Name].
- Implemented [model/algorithm] in PyTorch, achieving [X]% improvement on [benchmark].

EDUCATION

Kwame Nkrumah University of Science & Technology (KNUST)

B.Sc. Computer Science | Kumasi, Ghana | Expected 2029

- Relevant Coursework: Networking & Internet (OSI/TCP/IP, IPv4, subnetting), Calculus I, Probability & Statistics, C++ / CS & Math Programming

Prempeh College (Senior High School)

WASSCE — General Science | Kumasi, Ghana | Graduated 2025

PROJECTS

[Computer Vision Project — e.g. Crop Disease Detector]

[PyTorch · CNN/ResNet · FastAPI · AWS] | [github.com/...] [live-demo]

- Trained [CNN/ResNet/YOLO] on [X]K images achieving [X]% accuracy on [dataset]. Solved [real-world problem].
- Built inference pipeline: image upload → preprocessing → prediction → confidence score in [X]ms.
- Deployed via FastAPI on [AWS/GCP], serving [X] predictions/day. THIS is what Anthropic/Google want to see.

[NLP / LLM Project — e.g. AI Study Assistant]

[Claude/OpenAI API · LangChain · RAG · Next.js] | [github.com/...] [app-url]

- Built RAG pipeline: document ingestion → embedding → vector search → LLM generation with [X]% relevance.
- Implemented prompt chaining and guardrails for safe, grounded responses. [X] queries/day.

[Predictive ML Project — e.g. Energy Demand Forecasting]

[Scikit-learn · XGBoost · Pandas · Streamlit] | [github.com/...] [live-demo]

- Built [regression/classification] model with [X]% accuracy. Feature engineering from [X] raw features.
- Deployed interactive dashboard on Streamlit for real-time predictions.

Arethos — Mental Wellness App (3-Person Team)

React Native · Expo SDK 54 · SQLite · Groq & Anthropic AI | [github.com/Oppong-py/arethos](#)

- Machine Learning & AI
- Robotics & Embedded Systems
- UI/UX & Visual Design
- Movies & Video Games
- Sports (Football)

- Co-built a cross-platform mental wellness app for KNUST students with AI-powered journaling, mood tracking, and reflections.
- Engineered dual-AI fallback (Groq + Anthropic), custom haptic feedback calibrated through 9 iterations, and offline-first SQLite storage.

Autonomous Line-Following Robot

LEGO MINDSTORM EV3 · PID Control Algorithm

- Designed PID control algorithm (a core ML precursor) for autonomous navigation; National Champion at Robofest Ghana 2025.
- Optimized sensor-motor feedback loops — directly applicable to reinforcement learning and robotics control.

Class Attendance Monitoring System

React Native · Django · PostgreSQL · Bluetooth

- Architected biometric attendance platform serving 50 students in prototype testing, reducing manual roll-call by ~80%.
- Built Django REST API handling concurrent Bluetooth fingerprint scans — experience in data ingestion at scale.

Solar-Powered IoT Watering System

Arduino · IR Sensor · C++ · Solar

- Designed sensor-driven embedded system with proximity-triggered actuation — edge computing for IoT.
- Engineered full circuit with solar charge controller, relay logic, and sensor fusion.

◆ HACKATHONS & COMPETITIONS

National Cybersecurity Challenge — Regional Champion (2024)

- Competed in a knowledge-based quiz testing cybersecurity fundamentals, threat analysis, and security best practices.

Robofest Ghana — National Champion — Robotics (2025)

- Built autonomous robots under timed conditions to complete designated tasks; designed and assembled within strict time limits.

ACES CodeFest 2026 — UI/UX Challenge — Upcoming (July 2026)

- Competing in the ACES KNUST CodeFest UI/UX challenge — redesigning the ACES website into a mobile-first experience across 9 pages in a 2-week sprint.

◆ LEADERSHIP & VOLUNTEERING

Club Leader — Prempeh College Cybersecurity Club (2023–2025)

Participant — International Public Policy Forum (IPPF) (2024)

Online Safety Analyst & Volunteer — Solutions for Life Initiative Ghana (SFLIG) (Oct 2022–Present)

- Joined as volunteer (Oct 2022) for community support projects; promoted to Online Safety Analyst (Aug 2023), managing social media and website with a focus on internet security.
- Runs speaking engagements with girls from orphanages and rural communities on online safety topics.

◆ MEMBERSHIPS & COMMUNITIES

Member — Prempeh College Robotics Club (2023–2025)

Member — Association of Computer Engineering Students (ACES), KNUST (2023–Present)

Member — AmaliTech Coding Club (2025–Present)

Member — Microsoft User Group Ghana — KNUST Chapter (2025–Present)

Member — Claude Builder Club — KNUST (2025–Present)

References available upon request